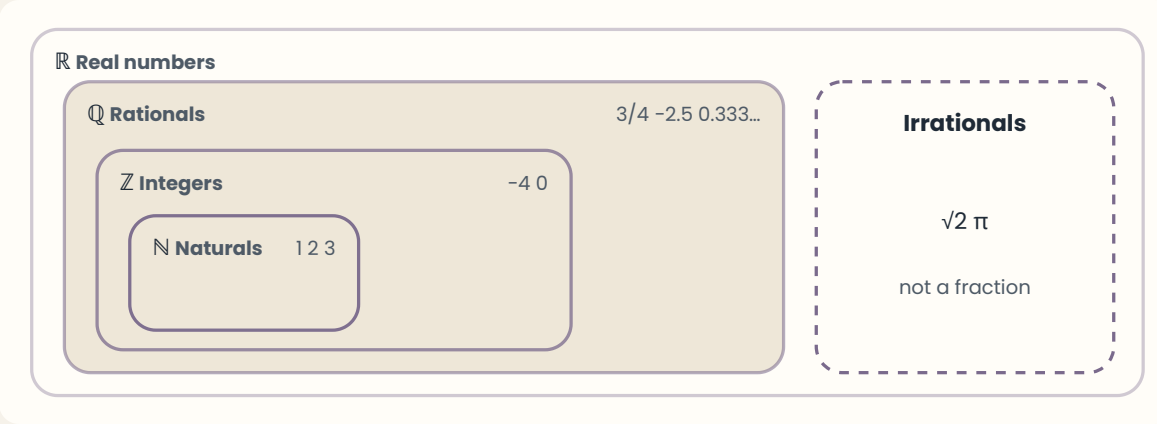
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Calculus 1 · Week 1

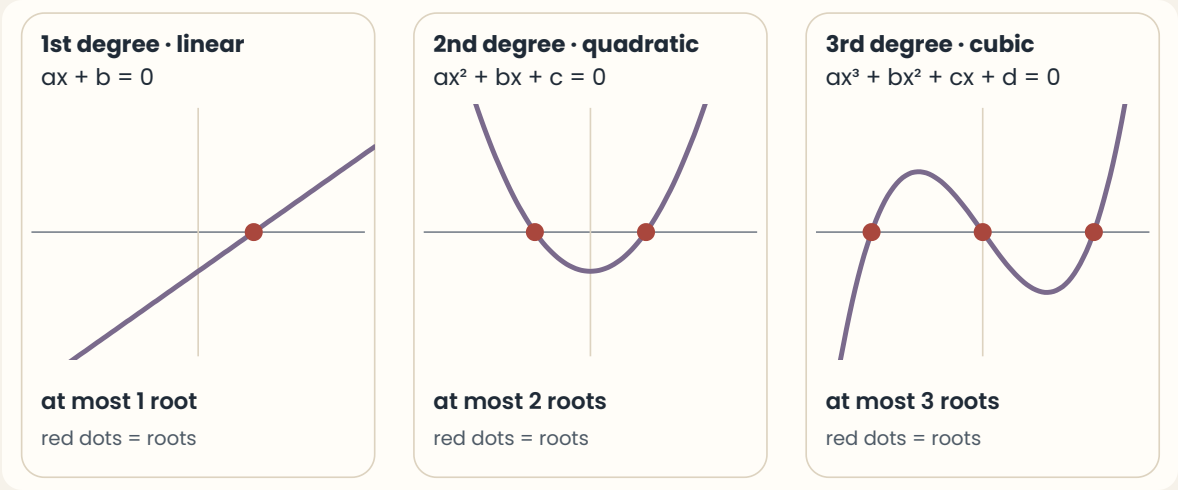
Introduction and inequalities

$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$, and the irrationals complete $\mathbb{R}$.



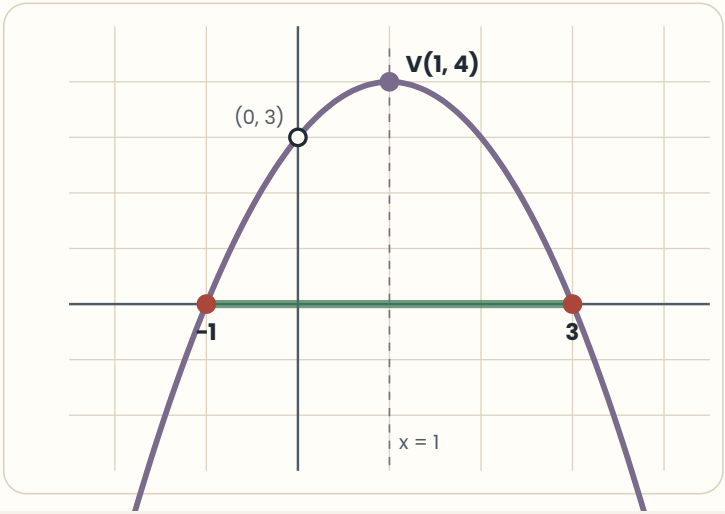
> **Think of it like this.** Think of the number line as a long street where every address exists, even 2.5 and √2. "Less than" just means "further west", and a set is a list of the houses that follow a rule.

Degree = highest power after expanding, moving everything to one side, and simplifying.



> **Think of it like this.** An equation is a balance scale: whatever you do to one pan, you do to the other, and it stays level. The degree tells you how many times, at most, the graph can cross the floor (the x-axis): a line once, a parabola twice, a cubic three times.

$x = (-b \pm \sqrt{\Delta})/2a$ with $\Delta = b^2 - 4ac$: $\Delta > 0$ two roots, $\Delta = 0$ one, $\Delta < 0$ none.



$y = -x^2 + 2x + 3$

$a = -1 < 0$: opens down

Vertex = maximum
$V = (-b/2a, -\Delta/4a)$
$= (-2/(-2), -16/(-4))$
$= (1, 4)$

Axis: $x = 1$

Roots: -1 and 3 ($\Delta = 16 > 0$)

Green: $y > 0$ between the roots

y-intercept: $c = 3$

> **Think of it like this.** A parabola with $a < 0$ is the path of a ball thrown in the air: it rises, stops for an instant at the top (the vertex), and comes down the same way it went up. The axis of symmetry is the mirror between the two halves.

Radicals: isolate, square, then check every answer.

- $|x| = x$ if $x \geq 0$, $-x$ if $x < 0$; $|A| = k \rightarrow A = k$ or $A = -k$.
- Fractions: write the forbidden values first, then multiply by the denominator.

> **Think of it like this.** Squaring both sides is like photocopying a document: the copy can carry marks the original did not have. Always compare the answer with the original.

Name the unknown with its unit before writing anything else.

- Translate sentence by sentence; check the answer in the story.
- Discard answers that make no sense: negative lengths, fractions of people.

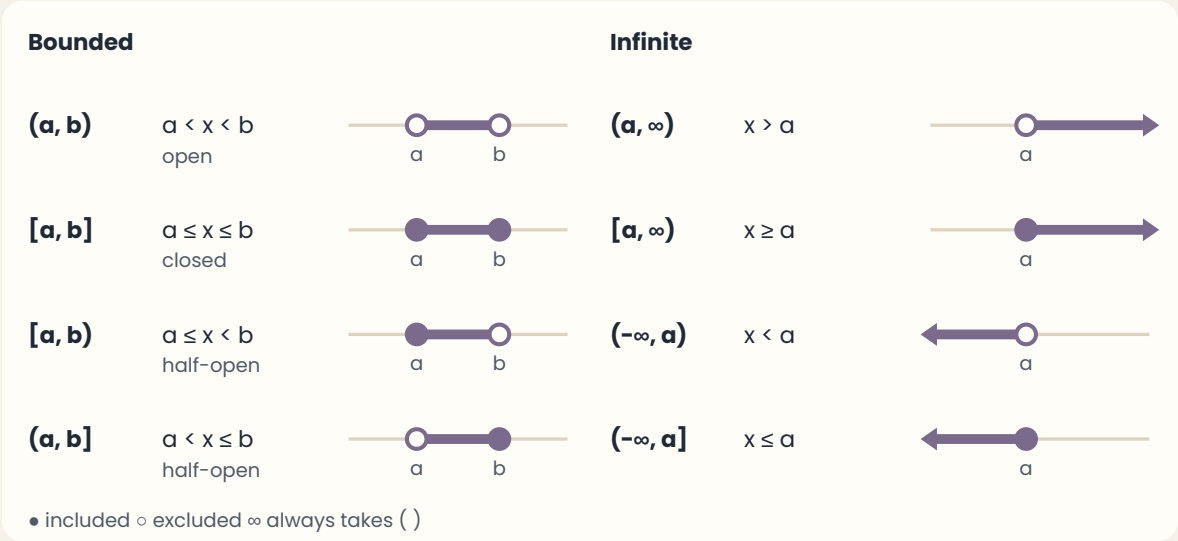
> **Think of it like this.** Translating a problem is like following a recipe written in another language: name each ingredient (the unknowns) before you start cooking (solving).

Candidates: ± divisors of the constant (over divisors of the leading coefficient).

- Remainder 0 means a root; the bottom row is the quotient.
- Write 0 for missing powers, and repeat until a quadratic is left.

> **Think of it like this.** Ruffini's rule is like peeling an onion: each root you find removes one layer, until the core is a quadratic you already know how to solve.

Bounded: (a, b), [a, b], [a, b), (a, b]. Infinite: (a, ∞), [a, ∞), (−∞, a), (−∞, a].



> **Think of it like this.** A bracket is a door you can stand in; a parenthesis is a door you can walk up to but never step through. Multiplying by a negative is a mirror: left and right swap, so "smaller" becomes "larger".

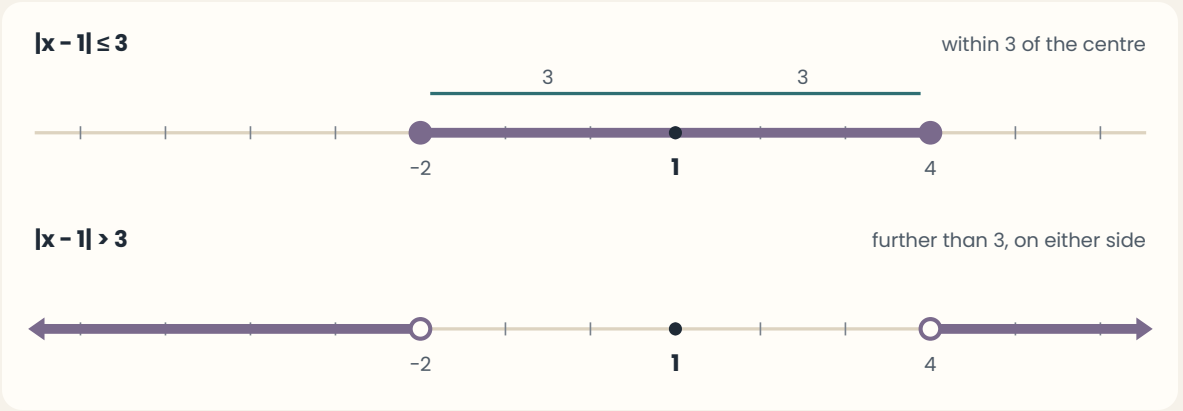
Zero on one side, factor, critical points, one test value per interval.

Interval	$x < -4$	$-4 < x < 3$	$x > 3$
Test value	$x = -5$	$x = 0$	$x = 4$
$x + 4$	−	+	+
$x - 3$	−	−	+
$(x + 4)(x - 3)$	+	−	+



Think of it like this. Critical points are traffic lights on the number line. Between two lights the colour cannot change, so one look per block is enough.

A system keeps the intersection of the solutions.



Think of it like this. |x − a| < r is a dog on a leash of length r tied to a post at a: it can be anywhere within r of the post, on either side. A system is a group of friends choosing a restaurant: only places everyone accepts make the list.

Step by step

Solve a cubic

1. List candidates: ±1, ±2, ±3, ±6
2. Try $x = 1$: $1 - 6 + 11 - 6 = 0$
3. Ruffini by $(x - 1)$: quotient $x^2 - 5x + 6$
4. Factor the quotient: $(x - 2)(x - 3)$
5. Roots: 1, 2, 3

Check yourself

1. How many real roots has $4x^2 - 4x + 1 = 0$?

A) Two B) One (repeated) C) None

2. A drone's height is $h(t) = -4t^2 + 24t + 5$ metres. What is its maximum height?

A) 41 m, at t = 3 s B) 5 m, at t = 0 s C) 77 m, at t = 6 s

3. A classmate solved $x^2 = 16$ and wrote "x = 4". What is missing?

A) The root x = -4 B) Nothing C) The root x = 8

Answers are in the speaker notes.

Practise next

Labs

- Equations of every type
- The parabola and Ruffini's rule
- Inequalities and sign tables
- Systems, cases, parameters, and regions

On the site

- Graph Lab
- Ruffini Lab
- Inequality Lab